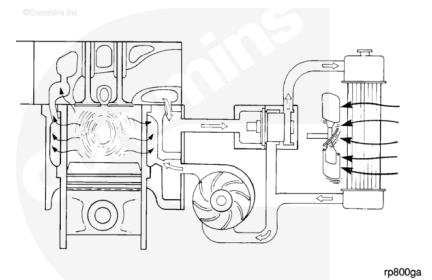


Trainings ▾

General Information

with Mechanically Actuated Injector

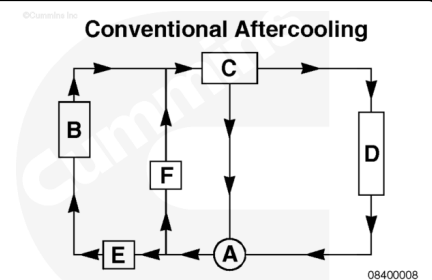
The primary function of the cooling system is to remove heat energy, created by the combustion process, from the engine. The excess heat energy that is **not** removed by the cooling system is carried away by exhaust gases and radiation into the atmosphere.



The accompanying chart illustrates the Conventional Aftercooling coolant flow through the engine.

For more details, Refer to Procedure 200-003 in Section (</qs3/pubsys2/xml/en/procedures/20/20-200-003-tr.html>)

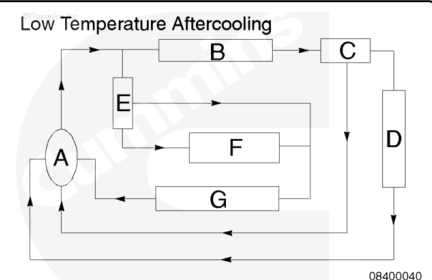
- Engine Cylinder Head
- Thermostat Housing
- Radiator
- Pump
- Aftercooler
- Oil Cooler.



The accompanying chart illustrates the Low Temperature Aftercooling coolant flow through the engine.

For more details, Refer to Procedure 200-003 in Section (</qs3/pubsys2/xml/en/procedures/20/20-200-003-tr.html>)

- Water Pump
- Base Engine
- Thermostat, Engine
- Radiator, Engine
- Low Temperature Aftercooling Thermostat
- Low Temperature Aftercooling Radiator
- Aftercooler



with Electronically Actuated Injector

The QSK19 CM850 engine maintains the standard 1-pump, 1-loop engine cooling system that requires both a radiator and air-to-air charge air cooler.

Heat rejection to coolant has been slightly reduced as a result of lower intake air temperatures and increased air flow. Maximum top tank temperature, however, remains unchanged at 212 °F [100 °C]. The torque converter cooler option remains the same and the corrosion filter continues to be required and also remains unchanged. The service interval of the corrosion filter is 250 hours.

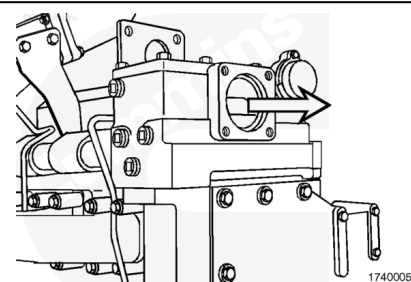
The coolant inlet and outlet connections are the same size and are located in the same location as the Tier 2, QSK19 engine configuration. As such, the same water inlet and water outlet options will be offered.

The turbocharged QSK19 Tier 3 engine requires the use of an air-to-air heat exchanger to reduce the temperature of the air entering the intake manifold. This type of heat exchanger is termed an air-to-air charge air cooler (CAC). The charge air cooler **must** be designed to meet the new intake manifold temperature target of 49 °C [120 °F] on a 25 °C [77 °F] ambient day.

The charge air cooling system includes the ducting from the turbocharger compressor outlet to the charge air cooler air inlet and from the charge air cooler air outlet to the engine intake manifold. The charge air cooler core and plumbing **must** be designed to meet the temperature and pressure requirements specified in the appropriate data sheets. It **must** also be resistant to corrosion and vibration (See AEB 24.06.).

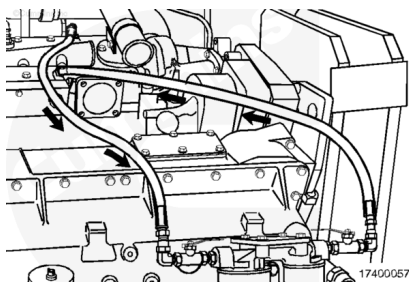
Rail Applications

Because the QSK19 Rail engine uses an air-to-air aftercooler, there is **no** aftercooler element in the intake manifold that uses coolant to cool the intake air. The thermostat housing has been modified to allow for coolant to flow to the OEM supplied radiator. The thermostat housing also contains a plug for the coolant bypass port. The plug is held in place by a tube clamp.

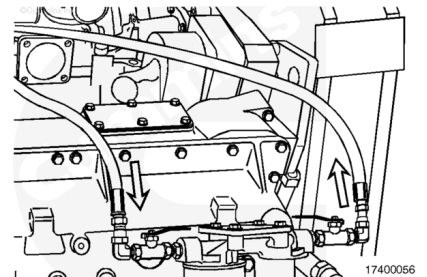


Note : The thermostat housing does not have thermostats. The coolant temperatures is controlled by the OEM supplied thermostats.

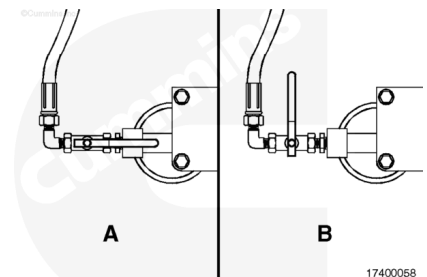
The coolant filter head is remote mounted to the side of the lubricating oil pan by a bracket. Plumbing for the unfiltered/treated coolant is returned to the engine through an adapter on the back of the water pump.



Two shut-off valves are located on either side of the coolant filter head on the inlet and outlet elbows.

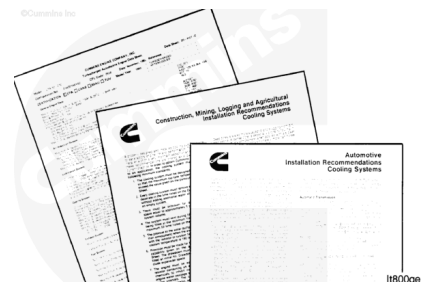


The two shut-off valves have an open (A) and a closed (B) position.



The following publications, available through a Cummins® Authorized Repair Location, provide cooling system installation recommendations and specifications approved by Cummins® Inc.:

- Data Sheets for Construction, Mining, Logging, and Agriculture, Bulletin 3381194; for Generator Drive and Generator Set, Bulletin 3381174; for Automotive, Bulletin 3381237.
- Operation of Diesel Engines in Cold Climates, Bulletin 3379009.
- Generator Drive and Generator Set Installation Recommendations, Bulletin 3382395.
- Coolant Requirements and Maintenance, Bulletin 3666132.



Refer to Procedure 205-002 in Section
 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
 (Service Literature Ordering Location) in section L for literature
 ordering information.

Marine Applications

Heat Exchanger

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The marine heat exchanger assembly is attached to the expansion tank

The marine heat exchanger is a plate or tube, tube design and is located on the front side of the engine. The plates are removable for cleaning. A sacrificial plug is located in the front cavity of the heat exchanger sea water passage.

The engine coolant flows from the thermostat housing outlet into the top rear of the heat exchanger and is directed over the plates or tubes. The engine coolant exits the cooler at the bottom rear of the heat exchanger. The engine coolant is then returned to the engine coolant pump inlet via a transfer tube.

The cooling sea water enters the bottom rear of the heat exchanger. The sea water flows through the tubes to the front, nondivided cavity. The sea water then flows through the tubes or plates and is discharged out of the top side of the heat exchanger. The sea water is then discharged overboard.

Marine Keel Cooling

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The marine single loop LTA cooling flow and engine temperature is controlled with one thermostat located in the thermostat housing. The engine coolant flows to the vessel mounted keel coolers through flexible couplings located at the front of the engine. The water pump inlet and outlet connections have been designed with a second connection for emergency inlet and outlet coolant connections.

The thermostat housing contains diagnostic ports for troubleshooting the keel coolers and after cooler.